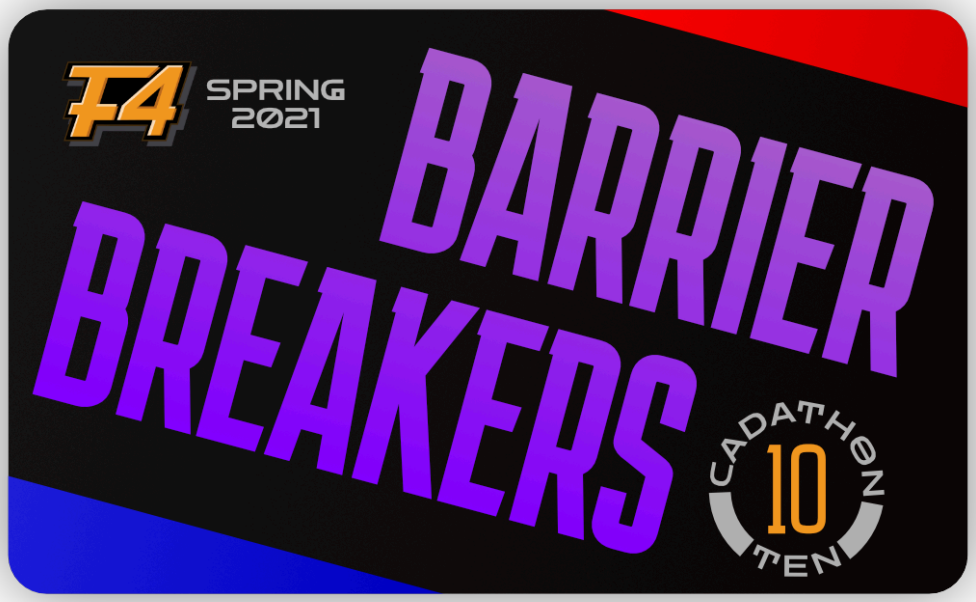


10th Bi-Annual F4 CADathon

2021 Spring

Section 1: Introduction

The intent of this manual is that the text means exactly, and only, what it says. Please avoid interpreting the text based on assumptions about intent, implementation of past rules, or how a situation might be in “real life.” There are no hidden requirements or restrictions. If you’ve read everything, you know everything.

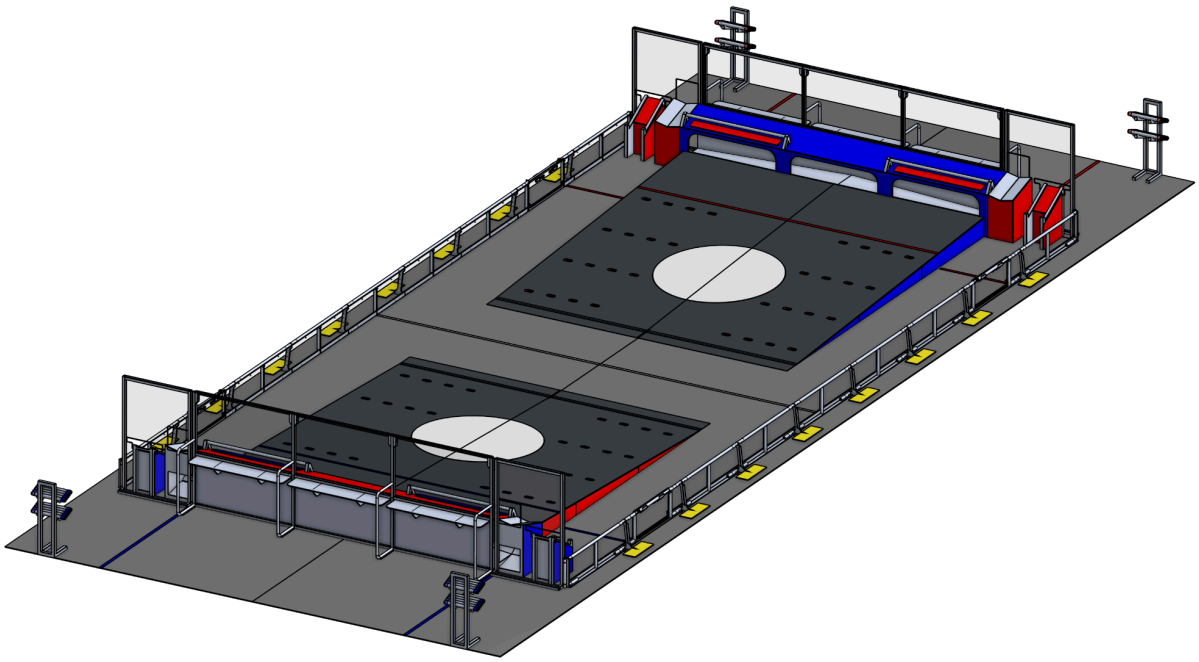
The rule numbering scheme uses an indication of the section in which the rule is stated plus a serial numbering system (e.g. safety rules begin with “S,” game rules begin with “G,” etc.). References to specific rules use this scheme

If you have a question regarding this game manual, the Game Design Committee (GDC) will answer questions on the Official Q/A. To submit a question go **here**. To view previously asked questions and the answers to said questions, go **here**. The Q&A is intended to help clarify rules, and sometimes the responses result in revisions to the text in the official document (which is communicated using Team Updates)

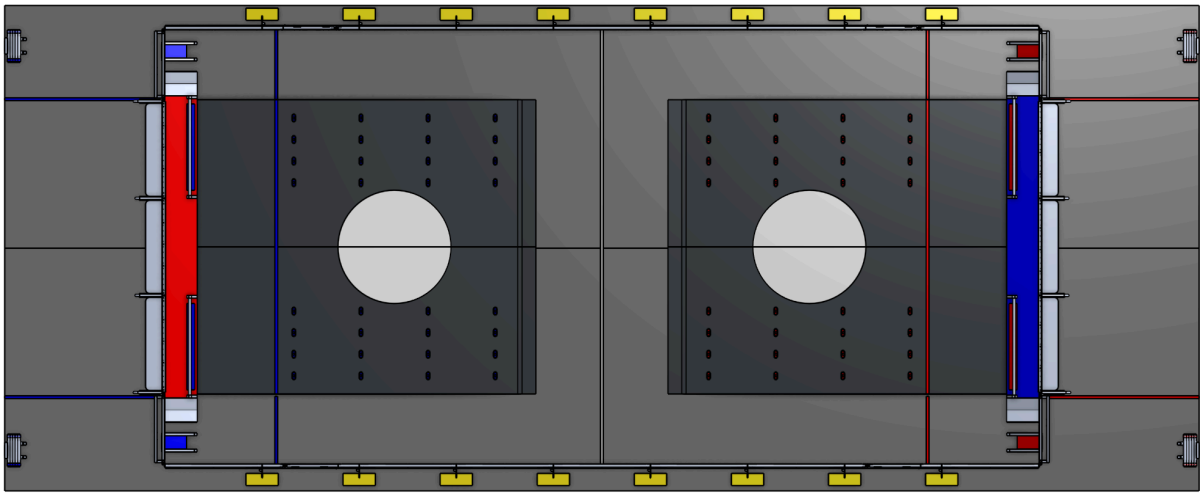
The GDC may release team updates throughout the course of the competition to notify teams of changes to the game manual. Updates will be posted in the Discord announcements channel.

Section 2: Overview

For the 10th F4 CADathon we will be playing *BARRIER BREAKERS*. *BARRIER BREAKERS* is a game played by two (2) ALLIANCES of up to three (3) ROBOTS on a 27’ by 54’ carpeted FIELD. *BARRIER BREAKERS* climbing the RAMP, BREAKING and ASSEMBLING BARRIERS, ACTIVATING the SUPERCHARGER to score GOALS, and CLIMBING using the CURL UP BAR.



Section 3: Arena



3.1: FIELD Layout

- **ALLIANCE WALL**
- **HUMAN PLAYER STATION**
- **GOAL**
- **RAMP**

3.2: FIELD

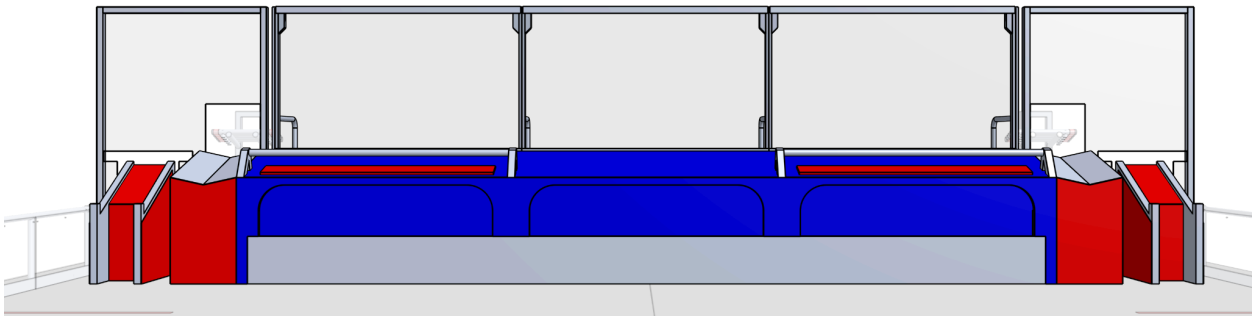
The FIELD for *BARRIER BREAKERS* is a 27 ft. (~823 cm) by 54 ft. (~1646 cm) carpeted area, bound by and including the inward-facing surfaces of the GUARDRAILS, RAMP, and ALLIANCE WALLS. The carpet used for the FIELD is gray in color (Shaw Floors, Philadelphia Commercial, Neyland II 20, “Ground Pepper”).

3.3 Guardrail

The GUARDRAIL is a system that consists of transparent polycarbonate supported on the top and bottom by aluminum extrusion. The GUARDRAIL prevents ROBOTS from inadvertently exiting the FIELD during a MATCH. There are four (4) gates in the GUARDRAIL that allow access to the FIELD for placement and removal of ROBOTS. The gates are 3 ft. 2 in. (~97 cm) wide and closed and shielded during the MATCH.

3.4 ALLIANCE WALL

The ALLIANCE WALL is the structure that separates ROBOTS from DRIVE TEAMS and consists of three (3) PLAYER STATIONS, an ALLIANCE-specific GOAL, and two (2) ALLIANCE-specific HUMAN PLAYER STATIONS.



3.4.1 PLAYER STATION

A PLAYER STATION is one (1) of three (3) assigned positions in an ALLIANCE WALL from where a DRIVE TEAM operates their ROBOT. Each PLAYER STATION is made from a 3 ft. (~91 cm) tall diamond plate panel base topped with a 3 ft. 6 in. (~107 cm) tall transparent plastic panel. An aluminum shelf is attached to each PLAYER STATION to support the DRIVE TEAM'S OPERATOR CONSOLE. The shelf is 5 ft. 9 in. (~175 cm) wide and 1 ft. (~30 cm) deep. There is a 4 ft. 6 in. (~137 cm) long by 2 in. (nominal) wide strip of hook-and-loop tape ("loop" side) along the center of the support shelf that may be used to secure the OPERATOR CONSOLE to the shelf

3.4.2 GOAL

The GOAL is composed of three 64" wide by 14" tall openings spaced equidistant across the Alliance Wall. The back of the GOAL is lined with 2" of open-cell foam in the alliance color, to reduce bounce-out.

Above the GOAL are two CURL-UP BARS. The CURL UP BARS consist of a 6’ (1.8m) length of 1.75” (4.5cm) diameter 6061-T6 Aluminum tube. Each CURL UP BAR is accompanied by a CONTACT PAD. The CURL UP BAR and CONTACT PAD are capable of handling any load that is applied to them in the course of playing *BARRIER BREAKERS*.

The CONTACT PAD is a 6” long by 64” wide piece of ¾” thick orange peel HDPE.

3.4.3 HUMAN PLAYER STATIONS

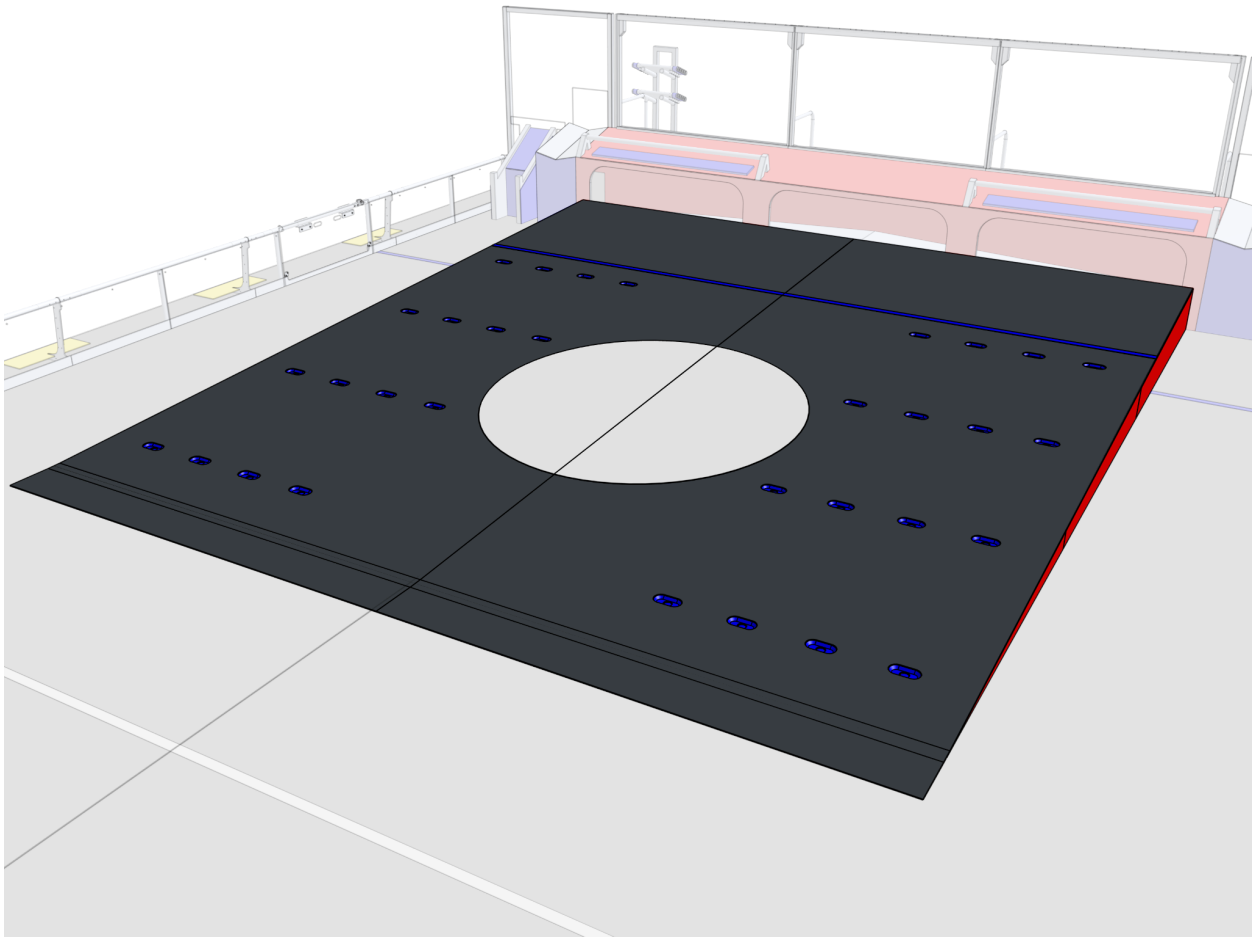
The HUMAN PLAYER STATIONS are located on each side of the PLAYER STATIONS and consists of a BARRIER LOADING STATION and a BALL LOADING STATION.

The BARRIER LOADING STATION extends 24” (63.5cm) from the ALLIANCE WALL and results in BARRIERS being 17” (43cm) off the ground.

The BALL LOADING STATION consists of a ramp that extends 24” (63.5cm) from the wall, ending 27” (68.6cm) above the ground.

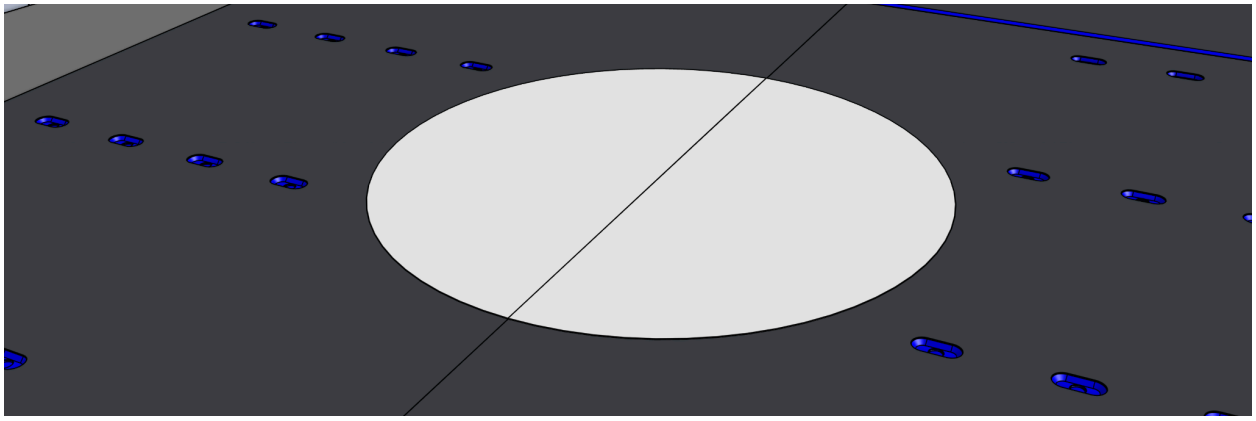
3.5 RAMP

Attached to the center of each ALLIANCE WALL is a RAMP with several features. The RAMP is an aluminum tubing structure covered with the same carpet used for the rest of the field (Shaw Floors, Philadelphia Commercial, Neyland II 20, “Ground Pepper”). The RAMP is 218.25” (5.5m) wide by 240” (6.1m) long by 13” (.33m) tall. It has an approximately 3 degree incline. The top of the RAMP contains the GOALIE ZONE. On the ramp is the alliance SUPERCHARGER and 16 BARRIER FOUNDATIONS.



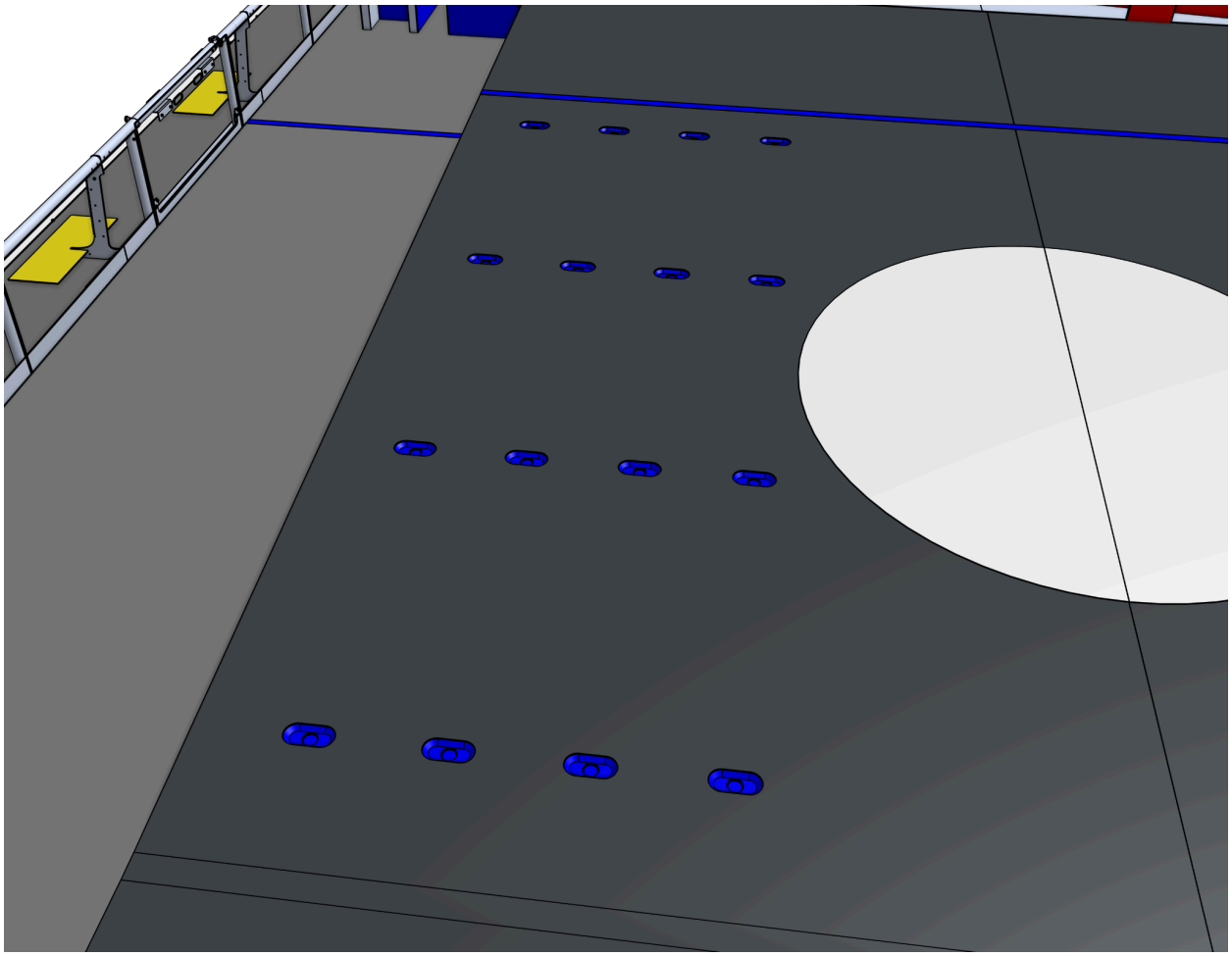
3.5.1 SUPERCHARGER

The SUPERCHARGER is a 7’ (2.13m) diameter circle of polycarbonate split into two HALVES. The SUPERCHARGER is ACTIVE when greater than 110lbs (50kg) of weight is present on both SUPERCHARGER HALVES. A single ROBOT present on both HALVES will not ACTIVATE the SUPERCHARGER.



3.5.2 BARRIER FOUNDATIONS

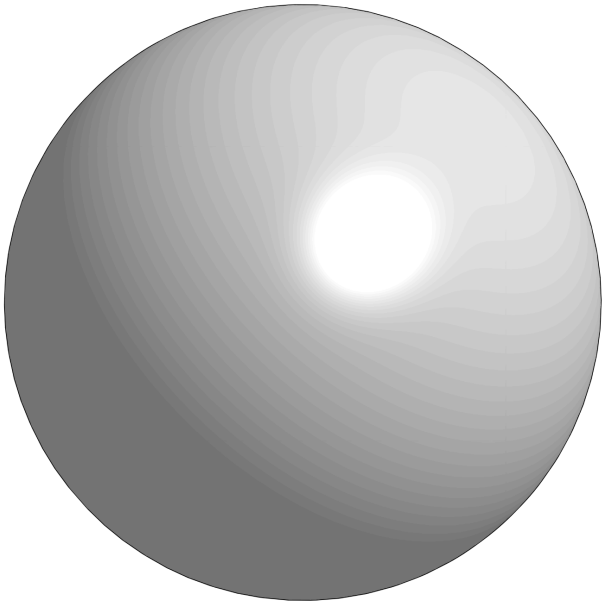
There are 16 BARRIER FOUNDATIONS present on each RAMP. BARRIER FOUNDATIONS consist of two 3” (7.5cm) diameter, 3” long slots in the RAMP with a depth of 1” (2.5cm) and a positive locking mechanism (Fasten Mate) that interacts with the BARRIERS. Once BARRIERS are ASSEMBLED, it will be impossible for them to move from this position for a 10 second period. A BARRIER is ASSEMBLED when both its legs enter the BARRIER FOUNDATION and the Fasten Mate takes effect.



3.6 Game Elements

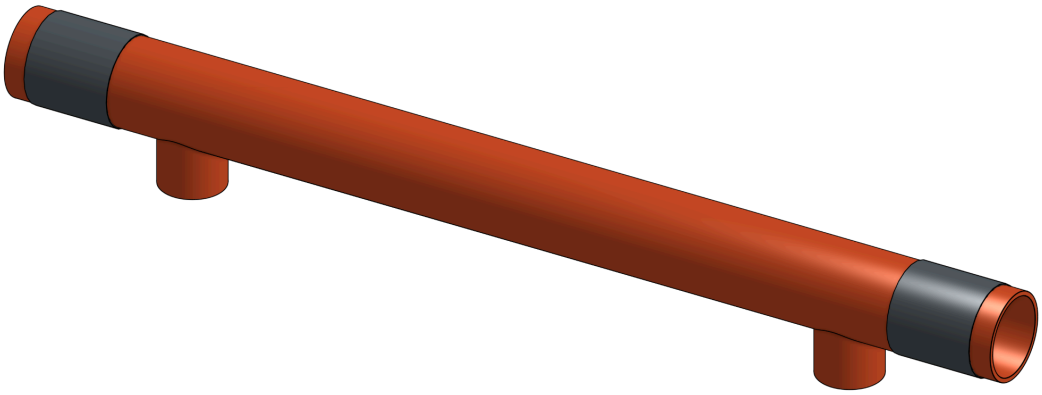
3.6.1 BALL

The BALL is a Size 5 Soccer Ball, weighing approximately 14-16 oz (410-450g), with a diameter of 8.66” (22cm).

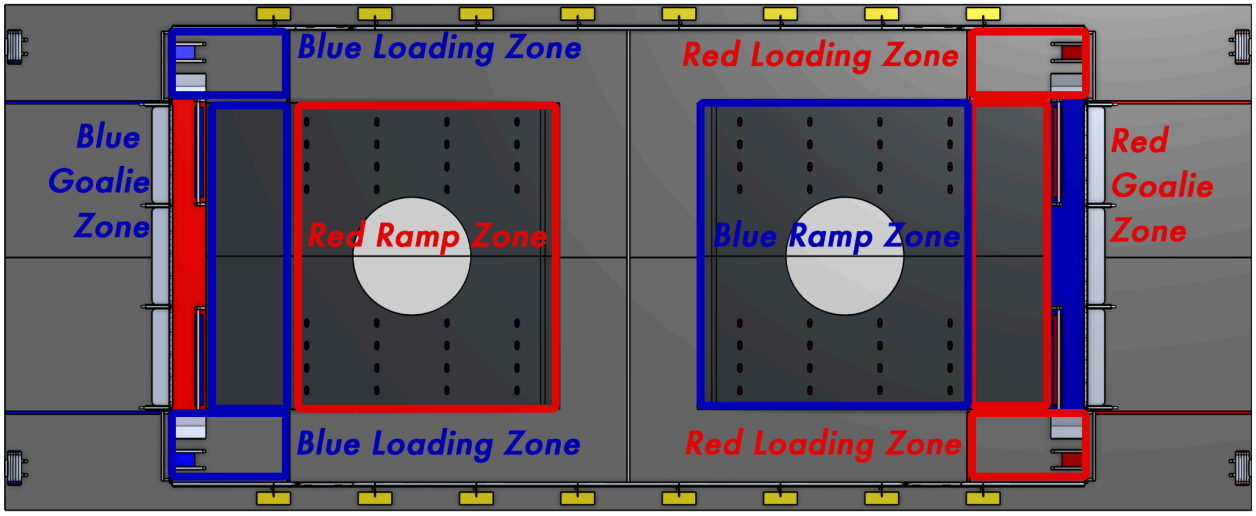


3.6.2 BARRIER

The BARRIER is a piece of 2” (5cm) diameter welded A500 Steel Pipe, with two 1.5” (3.8cm) diameter legs welded on. It is powder coated orange for superior performance. A BARRIER weighs approximately 4.6 lbs (2 kg).



3.8 Zones



3.9 Audio Cues

FMS alerts participants to milestones in the MATCH using audio cues. Please note that audio cues are intended to be a courtesy to participants and not intended as official MATCH markers. If there is a discrepancy between an audio cue and the FIELD timers, the FIELD timers are the authority.

- MATCH Start “Cavalry Charge” then Three (3) Bells
- T-30 seconds: Train Whistle
- T=0 for MATCH end: Buzzer
- MATCH stopped: Foghorn

Section 4: MATCH Play

4.1 MATCH Stages

MATCH Timing - Each MATCH is 2 minutes and 15 seconds long. There is no Autonomous period for the purpose of this event.

End Game - The end game takes place during the last 15 seconds of the MATCH.

4.2 MATCH Setup

At the beginning of the match 26 BARRIERS and 12 BALLS will be placed on the FIELD as follows:

- 7 BARRIERS in each ALLIANCE's RAMP ZONE, according to their direction (14 total)
- 3 BARRIERS in each of the HUMAN PLAYER STATIONS (12 total)
- 6 BALLS spaced equidistant on the CENTER LINE
- 1 BALL in each ROBOT's POSSESSION (6 total)

If a TEAM elects to not start the match with a BALL, that BALL will be placed in the possession of that ALLIANCE's HUMAN PLAYERs.

M01 When placed on the FIELD for a MATCH, each ROBOT must be:
A. in compliance with all ROBOT rules, i.e. has passed Inspection,
B. the only item left on the FIELD by the DRIVE TEAM,
C. confined to its STARTING CONFIGURATION,
D. contacting the SUPERCHARGER in their ALLIANCE's RAMP ZONE OR
E. fully contained in their ALLIANCE's GOALIE ZONE
Violation: If fix is a quick remedy, the MATCH won't start until all requirements are met. If it is not a quick remedy the offending ROBOT will be DISABLED and, at the discretion of the Head REFEREE, must be re-inspected.

M02 DRIVE TEAMS may not cause significant or repeated delays to the start of a MATCH and/or to the FIELD reset after the MATCH.
Violation: If prior to the MATCH, the offending DRIVE TEAM'S ROBOT will be DISABLED. If after the MATCH, YELLOW CARD.
DRIVE TEAMS are expected to stage their ROBOTS for a MATCH, and remove it from the FIELD afterwards, safely and swiftly. Examples include, but are not limited to:

A. late arrival to the FIELD
B. failing to exit the FIELD once the LED strings have turned off (indicating MATCH ready)
C. installing BUMPERS, charging pneumatic systems, or any other ROBOT maintenance, once on the FIELD
D. use of alignment devices that are external to the ROBOT (e.g. a DRIVE TEAM could bring and use a measuring tape, as long as there is no delay to the MATCH by doing so) E. failing to remove OPERATING CONSOLES from the PLAYER STATIONS in a timely manner

4.3 Scoring

When a BALL is SCORED it is returned via the opposing ALLIANCE's HUMAN PLAYER.

All endgame points are assessed and awarded 5 seconds after the MATCH ends.

4.3.1 MATCH Scoring

Action	Criteria	Points
BARRIER ASSEMBLED in ALLIANCE RAMP ZONE	BARRIER is ASSEMBLED at the end of the MATCH	5
BALL SCORED IN GOAL	BALL passes through the GOAL and enter the possession of the opposing HUMAN PLAYER	6
CLIMB	ROBOT is FULLY SUPPORTED by the ALLIANCE CURL UP BAR and/or CONTACT PAD	15
PARK	ROBOT is FULLY CONTAINED by their ALLIANCE's GOALIE ZONE	5
SUPERCHARGED GOAL	A BALL is SCORED when	9

	the ALLIANCE's SUPERCHARGER is ACTIVE	
Climb RP	35 or more ENDGAME points	1 Ranking Point
Barrier RP	All of the ALLIANCE's BARRIER FOUNDATIONS have a BARRIER ASSEMBLED	1 Ranking Point
Win the match	Score more points than your OPPONENT	2 Ranking Points
Tie the match	Score the same amount of points as your OPPONENT	1 Ranking Point

4.3.2 Rankings

Ranking Points (RP) are units credited to a team based on their ALLIANCE'S performance in Qualification MATCHES. Ranking Points are awarded to each eligible team at the completion of each Qualification MATCH per section 3.3.1 MATCH Scoring

Exceptions to Ranking Point assignment are as follows:

A. A SURROGATE receives zero (0) Ranking Points.
B. A DISQUALIFIED team, as determined by the Head REFEREE, receives zero (0) Ranking Points in a Qualification MATCH or causes their ALLIANCE to receive zero (0) MATCH points in a Playoff MATCH.
C. A "no-show" team is either DISQUALIFIED from or issued a RED CARD for that MATCH (see C6). A team is declared a no-show if no member of the DRIVE TEAM is in the ALLIANCE STATION at the start of the MATCH.

The total number of Ranking Points earned by a team throughout their Qualification MATCHES divided by the number of MATCHES they've been scheduled to play (minus any SURROGATE MATCH), then truncated to two (2) decimal places, is their Ranking Score (RS).

Order Sort	Criteria
1st	Ranking Score
2nd	Cumulative ENDGAME points
3rd	Cumulative BARRIER points
4th	Cumulative GOAL points
5th	Random sort by FMS

4.4: Rule Violations

Upon a rule violation, one or more of the penalties will be assessed and action will be taken according to the table below listed below

Action	Penalty
FOUL	5 points deducted from your score
TECH FOUL	15 points deducted from score
YELLOW CARD	A warning given by the REFEREES
RED CARD	The result of 2 YELLOW CARDS leading to the disqualification of a particular MATCH.
DISABLED	The ROBOT will be rendered inoperable for the remainder of the MATCH.

4.5 MATCH Logistics

M03 Any game piece that leaves the FIELD will be collected by FIELD STAFF and returned to MATCH play. Game pieces that inadvertently deflect back into the FIELD will be considered fair game. Note that ROBOTS may not deliberately cause game pieces to leave the FIELD (see G15), with the exception of scoring BALLS in the GOALS.

Section 5: Safety Rules

Safety is paramount at all times during any Tournament, and each rule below is intended to establish norms at each event that will mitigate injury risk to all participants

S01 ROBOTS whose operation or design is dangerous or unsafe are not permitted.
Violation: If before the MATCH, the offending ROBOT will not be allowed to participate in the MATCH. If during the MATCH, the offending ROBOT will be DISABLED.
Examples include, but are not limited to:
A. Uncontrolled motion that cannot be stopped by the DRIVE TEAM
B. ROBOT parts (or ROBOTS) that exit the playing FIELD
C. ROBOTS dragging their battery
D. ROBOTS that consistently extend beyond the FIELD
E. ROBOTS that pose risk of damage to the FIELD
F. ROBOTS that risk damage to other ROBOTS
G. ROBOTS that pose a danger to members of the DRIVE TEAM, spectators, or FIELD personnel

S02 ROBOTS and anything they control, e.g. a game piece, may not contact anything outside the FIELD.
Violation: Offending ROBOT will be DISABLED. Please be conscious of REFEREES, and FIELD STAFF working around the FIELD who may be in close proximity to your ROBOT.

Section 6: Game Rules

6.1 ROBOT Restrictions

G01 ROBOTS may not extend more than 12 in (30.48 cm). beyond their FRAME PERIMETER, except during the ENDGAME, when the ROBOT is in their ALLIANCE's GOALIE ZONE.
Violation: TECH FOUL.

G02 A ROBOT may not extend beyond their STARTING CONFIGURATION height.
Violation: TECH FOUL.

G03 ROBOTS may not intentionally detach or leave parts on the FIELD.
Violation: TECH FOUL. May escalate to DISABLED ROBOT, YELLOW CARD, RED CARD.

G04 ROBOTS must be in compliance with BUMPER rules throughout the MATCH.
Violation: FOUL. DISABLED if any BUMPER segment becomes completely detached, any side of a ROBOT is completely exposed (without BUMPER), or the Team number or ALLIANCE color becomes indeterminate.

G05 ROBOTS will not be re-enabled after the conclusion of the MATCH, nor will Teams be permitted to tether to the ROBOT except in special circumstances (e.g. during TIMEOUTS, after Opening Ceremonies, before an immediate MATCH replay, etc.) and with the express permission from the FTA or a REFEREE.

Violation: YELLOW CARD

6.2 ROBOT To ROBOT Interaction

G06 Strategies aimed at the destruction or inhibition of ROBOTS via attachment, damage, tipping, or entanglements are not allowed.

G07 Initiating deliberate or damaging contact with an opponent ROBOT on or inside the vertical extension of its FRAME PERIMETER, including transitively through a game piece, is not allowed.

G08 Two or more ROBOTS may not isolate or close off any major component of MATCH play, e.g. blocking GOALS or CURL UP BARS, shutting down all access to game pieces, quarantining all opponents to a small area of the FIELD, etc, and with the exception of robots present in their alliance's GOALIE ZONE

Violation: YELLOW CARD for the ALLIANCE. A single ROBOT blocking access to a particular area of the FIELD is not a violation of G08. Two ROBOTS independently playing defense on two opposing ROBOTS is not a violation of this rule. This rule has precedence over all other rules in this manual.

G09 Fallen (i.e. tipped over) ROBOTS attempting to right themselves (either by themselves or with assistance from a partner ROBOT) have one ten (10) second grace period in which they may not be contacted by an opponent ROBOT (even unintentionally). This protection lasts for either ten (10) seconds or until the protected ROBOT has completed the righting operation, whichever comes first.
Violation: FOUL. If intentional, YELLOW CARD.

G10 ROBOTS may not pin an opponent's ROBOT for more than five (5) seconds. A ROBOT will be considered pinned until the ROBOTS have separated by at least six (6) feet. The pinning ROBOT(s) must then wait for at least three (3) seconds before attempting to pin the same ROBOT again. Pinning is transitive through other objects. If the pinned ROBOT chases the pinning ROBOT upon retreat, the pinning ROBOT will not be penalized, and the pin will be considered complete.
Violation: FOUL, plus an additional FOUL for every five (5) seconds in which the situation is not corrected. If extended, RED CARD.

G11 Unless during the ENDGAME, or attempting to right a fallen (i.e. tipped over) ALLIANCE partner, ROBOTS may neither fully nor partially strategically support the weight of partner ROBOTS.
Violation: TECH FOUL.

G12 During ENDGAME, ROBOTS may not contact an OPPONENT'S ROBOT when any part of the OPPOSING ROBOT is contained within their GOALIE ZONE.
Violation: FOUL

6.3 FIELD Interactions

G13 DRIVE TEAMS, ROBOTS, and OPERATOR CONSOLES are prohibited from the following actions with regards to interaction with FIELD elements. Items A through B excuse DRIVE TEAM interaction with FIELD elements in their areas. Items C and D ..exclude climbing on the CLIMBING BARS. Item C excludes use of the PLAYER STATION hook-and-loop tape, plugging in to the provided power outlet, and plugging the provided Ethernet cable in to the OPERATOR CONSOLE.
A. Grabbing
B. Grasping
C. Attaching to (including the use of hook-and-loop tape against the FIELD carpet)
D. Hanging

E. Deforming
F. Becoming entangled
G. Damaging

Game Pieces are expected to undergo wear and tear. Minor scuffs, dings, and scratches are not a violation of G13.

G14 With the exception of Scoring, ROBOTS may not deliberately use game pieces in an attempt to ease or amplify the challenge associated with FIELD elements.
Violation: TECH FOUL per game piece. Repeated or egregious violations of this rule are likely to escalate rapidly to YELLOW or RED CARDS.

G16 ROBOTS may not CONTROL more than one (1) game piece at any given time.
Violation: TECH FOUL per Game Piece per instance.

G17 No more than one ROBOT may be in an ALLIANCE's GOALIE ZONE at a time, except during the ENDGAME
Violation: FOUL, if egregious or strategic, escalates to TECH FOUL

G18 A robot may not enter the opposing ALLIANCE's GOALIE ZONE
Violation: FOUL, if egregious or strategic, escalates to TECH FOUL

G19 A robot may only launch a BALL when the ROBOT is FULLY SUPPORTED (directly or transitively) by the alliance's RAMP, and the BALL is in contact with the ALLIANCE's RAMP.
Violation: FOUL, if egregious or strategic, escalates to TECH FOUL

The intent of G19 is that the BALL is in contact with the ground for the majority of the time the ROBOT is launching (imparting force on) the BALL. Judges will be advised that momentary or inconsequential violations of G19 are not to be penalized.

G20 A robot may not enter the opposing ALLIANCE's HUMAN PLAYER ZONE
Violation: FOUL, if egregious or strategic, escalates to TECH FOUL

G21 Only 3 BALLS maximum may be present in each HUMAN PLAYER STATION at a time.

Violation: FOUL, if egregious or strategic, escalates to TECH FOUL

Section 7: ROBOT Rules

The rules listed below explicitly address legal parts and materials and how those parts and materials may be used on a CADathon ROBOT. There are many reasons for the structure of the rules, including safety, reliability, parity, creation of a reasonable design challenge, adherence to professional standards, and impact on the competition.

F4 CADathon can be a full-contact competition and may include rigorous game play. While the rules aim to limit severe damage to ROBOTS, Teams should design their ROBOTS to be robust.

7.1: General ROBOT Design

R01 The ROBOT (excluding BUMPERS) must have a FRAME PERIMETER, contained within the BUMPER ZONE, that consists of fixed, non-articulated structural elements of the ROBOT. Minor protrusions no greater than ¼ in. (~6.3 mm) such as bolt heads, fastener ends, weld beads, and rivets are not considered part of the FRAME PERIMETER.

R02 In the STARTING CONFIGURATION (the physical configuration in which a ROBOT starts a MATCH), no part of the ROBOT shall extend outside the vertical projection of the FRAME PERIMETER, with the exception of its BUMPERS and minor protrusions such as bolt heads, fastener ends, rivets, cable ties, etc.

R03 In the STARTING CONFIGURATION, the maximum ROBOT size (excluding BUMPERS) must be constrained to a FRAME PERIMETER of 120 in. Minor protrusions permitted in R01 and R02 (that are less than ¼ in. (~6.3 mm) such as bolt heads, fastener ends, weld beads, and rivets) are exempt from the 120 in. limit.

R04 In STARTING CONFIGURATION, the maximum ROBOT height is 36 in. tall.

R05 ROBOTS may not extend more than 12 in. (~40 cm) beyond their FRAME PERIMETER except during the ENDGAME, when the ROBOT is in their ALLIANCE's GOALIE ZONE (see G01).

R06 The ROBOT weight must not exceed 125 lbs. When determining weight, the basic ROBOT structure and all elements of all additional MECHANISMS that might be used

in different configurations of the ROBOT shall be weighed together. For the purposes of determining compliance with the weight limitations, the following items are excluded:
A. ROBOT BUMPERS
B. ROBOT battery and its associated half of the Anderson cable quick connect/disconnect pair
(including no more than 12 in. (~30 cm) of cable per leg, the associated cable lugs, connecting bolts, and insulation)
C. tags used for location detection systems if provided by the event

7.2 ROBOT Safety And Damage Prevention

R07 Traction devices must not have surface features such as metal, sandpaper, hard plastic studs, cleats, hook-loop fasteners or similar attachments that could damage the FIELD. Traction devices include all parts of the ROBOT that are designed to transmit any propulsive and/or braking forces between the ROBOT and FIELD carpet

R08 Protrusions from the ROBOT and exposed surfaces on the ROBOT shall not pose hazards to the FIELD elements (including the game pieces) or people.

R09 ROBOT parts shall not be made from hazardous materials, be unsafe, cause an unsafe condition, or interfere with the operation of other ROBOTS. These include:
A. Shields, curtains, or any other devices or materials designed or used to obstruct or limit the vision of any DRIVERS and/or COACHES and/ or interfere with their ability to safely control their ROBOT

B. Speakers, sirens, air horns, or other audio devices that generate sound at a level sufficient to be a distraction

C. Any devices or decorations specifically intended to jam or interfere with the remote sensing capabilities of another ROBOT, including vision systems, acoustic range finders, sonars, infrared proximity detectors, etc.

D. Exposed lasers other than Class I.

E. Flammable gasses

F. Any device intended to produce flames or pyrotechnics

G.Hydraulic fluids or hydraulic items

H. Switches or contacts containing liquid mercury

I. Circuitry used to create voltages in excess of 24V

J. Any ballast not secured sufficiently, including loose ballast e.g. sand, ball bearings, etc., such that it may become loose during a MATCH.

K. Exposed, untreated hazardous materials (e.g. lead weights) used on the ROBOT. These materials may be permitted if painted, encapsulated or otherwise sealed to prevent contact. These materials may not be machined in any way at an event.

L. Tire sealant

R10 ROBOTS must allow removal of game pieces from the ROBOT and the ROBOT from FIELD elements while DISABLED and powered off.

7.3 ROBOT Design Requirements

R11 A Team must submit their ROBOT for Inspection. The ROBOT must be an electromechanical assembly designed by the CADathon Team to perform specific tasks when competing in *BARRIER BREAKERS*. The ROBOT should include all of the basic systems required to be an active participant in the game – power, communications, control, BUMPERS, and movement. The ROBOT implementation must obviously follow a design approach intended to play *BARRIER BREAKERS* (e.g. a box of unassembled parts placed on the FIELD, or a ROBOT designed to play a different game does not satisfy this definition). R11 requires that this ROBOT has been purpose-designed for *BARRIER BREAKERS.*

R12 All ROBOTS will go through a brief formal inspection by a GDC member. The inspection will go on during the judging process. If a team is breaking a rule it will be taken into consideration during judging and cause a 5 point penalty per broken rule.

R13 Although we cannot truly govern budget constraints we ask teams to use the honor system and follow the budget rules imposed by FRC for 2021. These rule changes can be found here: https://www.firstinspires.org/sites/default/files/uploads/resource_library/frc/game-and-season-info/RobotContstructionInspectionRuleUpdate.pdf. Remember, even though this is not an official FIRST event, we practice the values of Gracious Professionalism.

R14 Mechanical/electrical designs created before Kickoff are only permitted if the source files are from COTS part providers. If the part is designed by you, the part needs to be publically available or placed in the #pre-cad-release channel on our Discord server before and during the entirety of CADathon (see C01 for more information).

R15 Teams are required to CAD a single set of BUMPERS up to external specifications. This can be in any color or texture you like as long as it is PG. Adding a way to actually mount the BUMPERS is a good way to earn points with the JUDGES in terms of model detail. The BUMPERS will also objectively always be made of FRC regulation fabrics.

R16: Sections 9.5, 9.6, 9.7, 9.8, and 9.9 of the FIRST INFINITE RECHARGE manual must be followed as they reasonably pertain to F4 CADathon, except as modified in this manual, The manual can be found here: https://firstfrc.blob.core.windows.net/frc2020/Manual/2020FRCGameSeasonManual.pdf

7.4 Bumper Rules

R17 Except as allowed per G16, BUMPERS must be located entirely within the BUMPER ZONE, which is the volume contained between a virtual plane 10” (~25 cm) above the floor and a virtual horizontal plane 16in. (~40 cm) above the floor in reference to the ROBOT standing normally on a flat floor. BUMPERS do not have to be parallel to the floor.

Section 8: CADathon Rules

8.1 Logistics

C01 Premade CAD models may be used as long as they are of a COTS COMPONENT. If they are not a COTS COMPONENT they must be posted in the #pre-cad-release channel on our Discord server or be made available publicly before and during the CADathon. See R14

C02 There is no limit on CADathon team size IF everyone on the CADathon team is a member of the same FIRST (FRC etc.) team. For CADathon teams with members of different FIRST teams, teams may consist of up to three (3) members. Larger teams may be allowed at the discretion of the GDC.

C03 Communication between teams is allowed. Sharing parts however, is against the rules and will lead to the disqualification of both teams.

C04 Teams MUST submit the following

A. STEP or Parasolid File of the ROBOT assembly
   Native CAD files are appreciated but not required
B. Completed Scouting Document
C. At least 1 rendering of the entire ROBOT
   additional renderings are appreciated but not required

Please name all files and documents with your CADathon team number followed by a short description.

Ex. 0000_RobotSTEP, 0000-FullRender, 0000 Scouting Document, etc. We regularly have 10+ team files open at a time when organizing submissions, don't get lost in the cloud!

C05 The CADathon will begin at 6pm EST on 04/18/2021 and conclude at 8pm EST on 04/25/2021. All CAD files and Pit Scouting sheets must be submitted by 8PM on 04/25/2021. Renders may be submitted until 8pm 04/26/2021. More details can be found in the event folder.

Due to the extended event time period and amount of teams competing, late and incorrect submissions will not be tolerated. Give yourself plenty of time to submit your files and contact a GDC member in Discord or email F4CADathon@gmail.com if you have any questions.

C06 Have Fun, and if you think you see a hole in the rules tell us. No one will think you're clever if you exploit someone's accident as this is a learning experience for everyone involved. Please remember that although this CADathon is not an official FIRST event, we still practice Gracious Professionalism. Questions may be asked on the Q/A here and answers will be posted here. This is the only place to get an official answer or rule clarification. (the questions channel on Discord is no longer an official ruling by the GDC)

C07 All participants must accept the invitation to join the CADathon Discord chat that has been sent to the emails the participants registered with for the CADathon. The participants must also be active/periodically checking the Discord for game and rule updates and easy communication. If a participant believes they did not receive an invitation to the Discord, they should contact F4CADathon@gmail.com.

C08 Teams should refrain from sharing images of their robots until after the live show to allow the proper amount of "hype."

Teasers and other team promotions (build blogs, social media posts, etc.) are mostly allowed as long as full renders of the final robot are not posted. DM a GDC member to inquire about your potential promotion ideas.

C09 All submissions must be via the CADathon submission form. Details can be found in the event folder.

C10 Breaking design rules if not egregious will result in a 5 point penalty to the final judged score of a participant. Egregious rule breaking will result in a disqualification from the CADathon show.

C10 ROBOTS will be judged on the following criteria by a panel of JUDGES with design experience.

The Judging Rubric can be found here:
https://docs.google.com/spreadsheets/d/1ChIgrb9Uj6XRb66wMDGJWO7hQWBkzC1Xj-RMSpn71Vo/edit#gid=0



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